Are Coding Agents Ready for Feature Pull Requests? An Industrial Audit and Harness-Engineering Playbook

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Abstract

Industrial teams now ask a deployment question, not a benchmark question: *can current coding agents produce feature pull requests*—multi-file, API-changing, test-bearing changes—*that engineers would actually merge*? We answer this for Huawei's coding-agent rollout by auditing four commercial agents on 62 real feature-implementation tasks and reporting where they break. We treat each agent as a *harness* on top of a frontier model: Claude Code (Opus 4.6), Codex CLI (GPT-5.4), Cursor (Composer-2), and OpenCode (GLM-5.1). We use *harness* to mean the system around the model—planning, retrieval, tool use, edit policy, validation hooks, and review gates. The benchmark covers 62 tasks (12 from the Huawei CANN/`torch_npu` accelerator stack, MindSpeed, and Kubernetes workflows; 50 from our open-source extension CapBench, spanning Kafka, CPython, Airflow, TensorFlow, Django, and others) and a controlled long/short prompt comparison, yielding $4 \times 62 \times 2 = 496$ runs. Each run is scored by a four-evaluator LLM panel using a three-dimension production-equivalence rubric (functional correctness, completeness, behavioral equivalence) and a strict majority verdict. Across 496 runs, only 8 reach a production-equivalent verdict (1.6%); the strongest harnesses (Claude Code, Codex CLI) pass 2.4% each, the weaker harnesses (Cursor, OpenCode) pass under 1%. Every PASS run concentrates on four of the 62 tasks, and the entire Kubernetes track yields 0/32 PASS. The failure profile is mixed rather than a single "incomplete coverage" story: 41.4% of non-PASS runs are semantic/correctness failures, while 26.2% are coverage gaps and another 22.3% are borderline partials that expose incomplete file discovery, missing tests, or unfinished cross-module wiring. Long prompts lift the strongest harnesses from 0% to 4.8% PASS, while Cursor and OpenCode reach only 1.6%, suggesting that gains from richer specification are mediated by the harness, not the underlying model. We close with a four-step harness recipe (long-form task spec, scope contracts, output-side test/coverage gating, multi-evaluator acceptance) that follows directly from where today's agents fail in our audit. The takeaway for industry teams is concrete: ship coding agents now for low-coupling tasks, but invest harness engineering—not just larger models—to close the feature-PR gap.

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1 Introduction

Software engineering organizations are under growing pressure to turn coding agents from research demonstrations into deployable engineering infrastructure. The decision they face is not whether an agent can suggest code in an IDE, but whether it can produce a reviewable pull request for substantive feature work in a production codebase. At Huawei, this question is operational: multiple stack-level teams—Kubernetes, the CANN compiler stack (CANN), the MindSpeed training framework, and `torch_npu`—are deciding in 2026 whether to route portions of their feature backlog through coding agents. The deployment decision turns on a concrete question: *when an agent is given a feature specification and the corresponding repository, can it produce a candidate PR that a human reviewer would accept with at most light revision?*

Why existing benchmarks do not answer this question. Coding-agent benchmarks have made fast progress on bug-fix and short-task settings. Industry-watched leaderboards report that the strongest agent configurations now resolve over 70% of issues

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on SWE-bench Verified [13], and many leaderboards exist for short, well-localized tasks. But these tasks are predominantly bug repair: they target a small set of files identified in advance, often supply the failing test, and reward minimal local edits. Feature implementation differs in kind, not degree. A new feature requires the agent to identify the right architectural surface, modify several files in coordinated ways, add or update tests, respect repository-specific conventions, and produce a change set that integrates with parallel ongoing work. Industry surveys (e.g., [8]) confirm that feature work—rather than localized bug fixes—is the bottleneck developers most want help with. The gap between bug-repair benchmark scores and merge-ready feature pull requests is therefore not just a question of model size; it is a question about whether today's agents can deliver value on the part of the workflow that matters most to engineering teams.

An industrial audit. We answer the deployment question directly by auditing four leading commercial coding agents on a curated 62-task feature-implementation benchmark. The benchmark combines 12 *Huawei industrial tasks* drawn from the codebases that the rollout would touch (Kubernetes, the CANN/`torch_npu` accelerator stack, and MindSpeed) with 50 *open-source feature tasks* from a curated extension we call CapBench, drawn from 15 production-grade open source projects (Kafka, CPython, Airflow, TensorFlow, Apache Spark, Postgres, ClickHouse, etc.). Each task is paired with a *long* prompt (a multi-paragraph design proposal that mirrors a real feature spec) and a *short* prompt (a one-sentence ticket-style description), giving us $62 \times 2 = 124$ task instances per agent. We evaluate four agents—Claude Code (Opus 4.6), OpenAI Codex (GPT-5.4), Cursor Composer-2, and OpenCode/GLM-5.1—producing $124 \times 4 = 496$ end-to-end agent runs. Each run is independently scored on three production-relevant axes (functional correctness, completeness/coverage, and behavioral equivalence to the reference implementation) by an automated four-evaluator panel; the verdict is decided by 3-of-4 majority. We treat the four agents not as four versions of the same model but as four *harnesses*: each ships its own planning loop, tool integration, code-search retrieval, and verification logic, and these harness choices turn out to dominate the results.

This setup gives us a clean answer to the question Huawei's deployment teams asked: *can today's coding agents take over feature delivery for our internal codebases?*

Headline findings. The result is a clear boundary, not a flat yes or no:

- **Aggregate verdict: 8 of 496 runs pass strict majority review (1.6%).** Treating the four evaluators as a panel and requiring 3-of-4 PASS, only 8 runs reach production-equivalent quality. The remaining 488 runs are split across PARTIAL (194 runs, 39%) and FAIL (294 runs, 59%).
- **Harness gap dominates the model gap.** The two strongest harnesses (Claude Code and Codex CLI) tie at 2.4% PASS each; the two weaker harnesses (Cursor Agent and OpenCode CLI) sit at 0.8% PASS each—a 3× gap. The underlying base models (Claude Opus 4.6 and GPT-5 with reasoning) are state of the art; the harness layer (planning loop, retrieval policy, verification step, edit boundaries) explains most of the variance we observe.
- **Long-prompt advantage is asymmetric.** On long prompts (with full design context), the strong harnesses lift PASS rate from 0% to 4.8%; the weaker harnesses improve only to 1.6%. Better specification helps strong harnesses ship; for weak harnesses, more context improves scores but rarely converts outputs into production-equivalent PRs.
- **Kubernetes-class tasks are still out of reach.** All 4 harnesses produce 0/8 PASS on the four K-tasks (Kubernetes scheduler / kubelet feature work). Cross-cutting feature work in mature Go infrastructure remains the hardest deployment frontier we measured.
- **Failures are heterogeneous, but many are harness-addressable.** The 488 non-PASS runs split into semantic/correctness failures (41.4%), coverage gaps (26.2%), behavioral/design mismatches (10.0%), and borderline partials (22.3%). The important deployment signal is not that every failure is mere incompleteness, but that a large recoverable slice involves stopping early, omitting tests or generated artifacts, or failing to wire cross-module changes.

This negative result for autonomous feature delivery is also a positive result for harness engineering: **a meaningful share of the failures is systematically addressable even though semantic limits remain.** Coverage gaps plus borderline partials account for 48.5% of the non-PASS population, giving harnesses a concrete target: force scope discovery, test evidence, invariant checks, and cross-module closure before a draft reaches human review.

Contributions. This paper presents:

(1) An industrial benchmark of 62 feature implementation tasks (12 Huawei + 50 CapBench, an OSS extension we release alongside) with reference implementations and structured grading rubrics.
(2) A four-evaluator LLM-judging panel with majority-rule resolution, calibrated against expert reviewer agreement.
(3) Empirical results across 496 runs (4 agents × 62 tasks × 2 prompts) showing ≤2.4% PASS rate on the strongest harness and a categorical 0% PASS for Kubernetes feature work.
(4) A failure-mode taxonomy and four mapped harness recommendations (scope gating, file-enumeration plans, invariant assertions from spec, cross-module wire-up linting).

The remainder of the paper is structured as follows. Section 2 lays out the deployment question and the four task families that anchor it. Section 3 describes the 62-task benchmark, the four harnesses, the prompt and evaluation protocol, and the cheating-prevention guards. Section 4 reports the headline numbers (8/496 PASS), broken down by agent, family, and prompt. Sections 5 and 6 dig into why agents fail and what harness layer would catch it. Section 7 translates the results into deployment guidance for engineering teams considering coding agents for production work.

2 Industrial Context and Evaluation Scope

2.1 The Deployment Decision We Had to Make

The motivation for this study is an adoption decision rather than a leaderboard exercise. Huawei engineering teams working on accelerator operators, distributed training, and Kubernetes-based platform software are evaluating whether commercial coding agents can safely take on *feature-level* work, not just code completion or local bug fixes. The practical question is concrete: can an agent produce a first-draft pull request close enough to a senior engineer's implementation that a reviewer could land it with minor edits?

This deployment question decomposes into four research questions:

(1) **RQ1: Capability boundary.** Can current agents complete production-equivalent feature PRs without human intervention?
(2) **RQ2: Task-shape effect.** Does performance degrade gradually as feature size grows, or is there a cliff for cross-module work?
(3) **RQ3: Harness leverage.** How much can richer task specifications help without changing the underlying model or agent framework?
(4) **RQ4: Deployment policy.** Which task classes should be routed to agents today, and which should remain human-owned?

These questions shape the benchmark design, prompt variants, and deployment guidance reported in the rest of the paper.

2.2 Why Generic Benchmarks Were Insufficient

Before building our own evaluation, we considered existing coding-agent benchmarks. SWE-Bench [13] and its variants [5, 7] evaluate real repository patches, but the dominant workflow is bug fixing with an executable oracle. Feature implementation in our setting has a different acceptance condition: the agent must discover every file that belongs to the feature, add tests where appropriate, and follow the project's architectural conventions. We therefore needed an evaluation that:

- mirrored the *structure* of the work our engineers actually do (multi-file features, test additions, cross-subsystem refactors),
- drew tasks from *our* codebases so that language, API, and architecture matched deployment reality, and
- used a verdict rule robust to single-evaluator bias.

Existing public benchmarks cover parts of this space, but none provided all three properties simultaneously for our deployment decision.

2.3 Codebase and Task Scope

The benchmark pairs 12 tasks mined from Huawei-maintained production repositories with 50 tasks sampled from widely-used open-source repositories so that results generalise beyond one organisation's coding style.

Huawei and platform tasks (12 tasks). The 12 industrial tasks span three engineering areas selected with our engineering partners as representative consumers of an internal agent deployment:

- **CANN/`torch_npu` accelerator stack** (C1–C5): performance-sensitive C++/AscendC kernels and Python integration changes implementing new operators or refactoring existing ones; typical changes touch kernel code, host-side launch wrappers, operator registration tables, integration code, and unit tests.
- **MindSpeed training framework** (M1–M3): Python changes to the distributed-training runtime; typical tasks modify scheduling, memory accounting, or loss-scaling logic and require passing the MindSpeed integration suite.
- **Kubernetes platform** (K1–K4): feature work on the upstream open-source Kubernetes codebase (Go), including API server extensions, controller changes, and generator-driven conversion code spanning 10s of files per PR.

Open-source control set (50 tasks). The CapBench tasks (T01–T50) are merged PRs sampled from 15 widely-deployed open-source projects (Kafka, CPython, Airflow, Django, gRPC, Spring, Kubernetes OSS, and others). They provide an out-of-organization control set with similar multi-file feature shapes but without Huawei-specific APIs.

2.4 Task Curation Funnel

Raw PR archives are unsuitable as benchmark tasks because many PRs are trivial, non-reproducible, or depend on unstated team context. We therefore used the same four-stage curation funnel for both the Huawei and CapBench subsets:

(1) **Feature-work filter.** Retain PRs that implement a behaviour change, feature extension, or cross-file refactor; drop typo fixes, dependency bumps, formatting-only changes, and documentation-only changes.
(2) **Reproducibility filter.** Retain tasks whose base repository can be checked out and sanitised without requiring unreleasable ground-truth history, hidden remotes, or production credentials.
(3) **Reviewability filter.** Retain tasks whose expected outcome can be compared against a merged implementation with concrete files, tests, or behaviour, rather than an open-ended design discussion.
(4) **Diversity filter.** Stratify by repository family, language, file count, and diff size to avoid a benchmark dominated by one project or one task style.

The final benchmark contains 62 tasks: 12 Huawei/platform tasks and 50 CapBench tasks. We intentionally prefer a smaller, auditable benchmark over a large automatically harvested set because the study's purpose is deployment guidance, not training-data scale.

2.5 Deployment-Equivalence Criterion

We score agent outputs under a production-equivalence criterion: a PASS means the generated diff is close enough to the merged human PR that a reviewer could plausibly accept it after ordinary review. This criterion combines functional correctness, completeness of the edited scope, and behavioural equivalence to the accepted design. It is intentionally stricter than test-only resolution because industrial feature work often has multiple files, generated artifacts, and reviewer conventions that are not fully captured by a single

fail-to-pass test. Section 3 defines the evaluator panel and precise rubric.

2.6 Scope and Non-Goals

We do not measure productivity impact, developer satisfaction, or long-term maintenance outcomes; those require live deployment studies. We also do not study pair-programming loops with intermediate human feedback. Every run is fully autonomous because that is the strongest claim a procurement or platform team would need before routing feature work to an agent. Finally, we deliberately exclude UI/design tasks, documentation-only PRs, and PRs that were reverted or superseded.

3 Study Design

Figure 1 summarizes our experiment pipeline. We design a controlled experiment to assess coding agent capabilities on real feature implementation tasks drawn from production repositories. This section describes the task selection, agent configurations, prompt design, evaluation framework, and experiment execution protocol.

3.1 Task Selection

Our benchmark has 62 tasks in total: 12 from the Huawei/platform ecosystem (Kubernetes, the CANN/`torch_npu` accelerator stack, and MindSpeed) and 50 from *CapBench*, an open-source extension assembled from 15 widely used community projects (Kafka, CPython, Airflow, Terraform, Spring, etc.). Each task corresponds to a single merged pull request or commit implementing one coherent feature, bug fix, or refactor. Table 1 summarises the Huawei/platform subset; the full CapBench task inventory is included in the replication package.

Selection criteria. Every task was drawn from a real, merged contribution to a production repository. C-family accelerator tasks (C1–C5) and MindSpeed tasks (M1–M3) come from Huawei's Ascend NPU software stack—operator kernels, `torch_npu` integration, training framework features, and precision fixes that represent typical industrial workloads. Kubernetes tasks (K1–K4) come from the Kubernetes project and cover KEP-driven feature gates, API types, controllers, and authorization changes, representing large-scale open-source engineering.

Complexity classification. We classify tasks into three tiers based on the number of ground-truth files, cross-component coordination requirements, generated-code or API surface, and domain-specific knowledge needed. The thresholds are not pure file-count bins: a small Kubernetes API change can be High because it requires generated types, validation, tests, and release-cycle knowledge, while a bounded Python change can remain Medium even with a larger diff.

- *Low* (C1, M1): localised edits with limited architectural coupling.
- *Medium* (C2, C3, M2, M3): bounded multi-file changes or domain-specific implementation work that remains inside a narrow subsystem.
- *High* (C4, C5, K1–K4): cross-component changes that require architectural understanding, generated/API surface updates, multi-module orchestration, or extensive testing.

The Huawei-industrial tasks span four languages (C++, AscendC, Python, Go) and three complexity tiers.

CapBench extension (50 tasks). To broaden beyond the Huawei ecosystem, we added 50 tasks from CapBench, our open-source complementary suite curated from 15 widely-used projects: Kubernetes, Kafka, Spring Boot, CPython, Django, Airflow, TensorFlow, Apache Flink, LangChain, Kubernetes Operators, Redis, and others. Tasks are drawn from merged PRs that implement a real feature or fix, span 1–127 handwritten files, and cover six languages (Go, Python, Java, Rust, TypeScript, C++). CapBench tasks use the same metadata rubric: Low tasks are local edits, Medium tasks are bounded multi-file changes, and High tasks require broader cross-component coordination. Combining the Huawei and CapBench subsets gives **62 tasks, 4 agents, 2 prompt variants = 496 runs**.

3.2 Agent Configurations

We evaluate four primary agent configurations, summarised in Table 2. Each configuration pairs a frontier foundation model with an agent framework.

Product-level comparisons. Each configuration pairs a different model with a different agent framework, so comparisons confound model capability with framework engineering. These comparisons are best understood as *product-level* assessments—comparing the capabilities of commercially available agent tools—rather than controlled model ablations. We discuss this confound further in §9.

3.3 Prompt Design

Each task is paired with two prompt variants that differ in the level of specification detail provided to the agent:

- **Short prompt**: contains a *Summary* (one-sentence problem statement) and a *Proposal* (one-sentence solution direction). Typically 2–4 sentences.
- **Long prompt**: extends the short prompt with *Motivation* (why the change is needed) and *Design Details* (numbered list of implementation considerations). Typically 15–30 sentences.

Design principle. Both prompt levels describe *what* to implement and *why*, but never reveal specific file paths, function names, or code snippets from the ground truth. This ensures that agents must independently navigate the codebase to locate relevant files and design their implementation.

Example (C1, buffer alias fix). The short prompt reads: "*ScaledMaskedSoftmaxGradV2 has a tensor-lifetime bug: a buffer is used after its queue resource is freed. Replace the reference with an equivalent surviving buffer.*" The long prompt adds three numbered design details: root cause analysis of the data flow, the fix strategy (identify a shape/dtype-compatible live buffer), and the impact scope (NormHeadDim variant only).

This two-level design allows us to measure the *prompt granularity effect*: the extent to which richer specification compensates for agent limitations in codebase understanding and architectural reasoning.

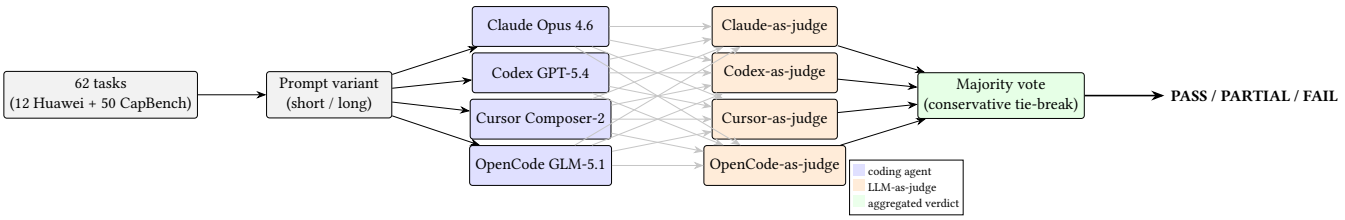

Figure 1: Evaluation pipeline. 62 tasks × 2 prompt variants × 4 agents yield 496 runs; each run is scored by four LLM judges (including the same model families as the agents themselves), then combined by majority vote with a conservative tie-break favouring PARTIAL over PASS. The pipeline produces 1,984 evaluator judgments in total.

Table 1: Huawei/platform task registry (12 of the 62 tasks used in this study). GT = ground truth (merged code). Complexity is the three-tier metadata used in the evaluation CSV and reflects file count, cross-component coordination, generated-code surface, and domain-specific knowledge. The remaining 50 tasks come from the CapBench open-source suite.

ID	Repository	Complexity	Lang	GT Files	GT Lines
C1	cann-ops-adv	Low	C++	1	3
M1	MindSpeed	Low	Python	3	19
C2	cann-ops	Medium	C++	4	9
C3	torch_npu	Medium	Python	9	637
M2	MindSpeed	Medium	Python	6	150
M3	MindSpeed	Medium	Python	10	948
C4	cann-ops	High	AscendC	24	1,273
C5	cann-ops	High	AscendC	27	3,372
K1	kubernetes	High	Go	13	1,010
K2	kubernetes	High	Go	35	3,759
K3	kubernetes	High	Go	49	11,000
K4	kubernetes	High	Go	98	6,507

Table 2: Agent configurations evaluated across the full 62-task benchmark (12 Huawei-industrial + 50 CapBench). Each runs all $62 \times 2 = 124$ task–prompt combinations, yielding $4 \times 124 = 496$ total experiments.

Config	Model	Framework	N
Claude	Claude Opus 4.6	Claude Code	124
Codex	GPT-5.4	Codex CLI	124
Cursor	Composer-2	Cursor Agent	124
OpenCode	GLM-5.1	OpenCode	124

3.4 Evaluation Framework

Ground truth. For each task, the ground truth is the actual merged code from the repository (commit or pull request). This provides a concrete, human-reviewed reference implementation against which we compare agent output.

Scoring dimensions. Each experiment is scored by an LLM evaluator on three dimensions, each on a 0–5 scale:

- **Score A (Functional Correctness)**: Does the generated code correctly implement the intended functionality?
- **Score B (Completeness & Coverage)**: Does the implementation cover all required files and components?
- **Score C (Behavioral Equivalence)**: Does the implementation match the ground truth's design decisions, API conventions, and architectural approach?

Verdict rules. We assign a three-level verdict based on the dimension scores:

- PASS: $A \geq 4$ **and** $B \geq 4$ **and** $C \geq 3$.
- FAIL: $A < 2$ **or** $B < 2$ **or** $C < 2$ (any dimension critically deficient).
- PARTIAL: all other cases.

File coverage metric. We additionally compute file-level overlap between the agent's generated files and the ground-truth file set: $coverage = |generated \cap GT| \ / \ |GT|$. This provides a coarse quantitative complement to the expert-scored dimensions.

Evaluation scope. Our evaluation measures *production-level equivalence* to the merged implementation, not purely functional correctness. An agent solution that correctly addresses the requirement but uses a materially different file organization, naming convention, or architecture than the accepted PR will score lower on Behavioral Equivalence (C) and file coverage. This design choice reflects our industrial interest: we assess whether agent output could replace a human-authored PR draft, not merely whether it contains a plausible local fix. As a consequence, our FAIL verdicts should be read as "not production-equivalent" rather than "functionally

impossible." Our root-cause analysis (§5) confirms this: many non-PASS cases stem from engineering completeness gaps rather than total semantic misunderstanding.

Multi-evaluator cross-validation. To mitigate evaluator bias, every experiment is independently scored by four LLM evaluators from different model families: Claude Opus 4.6, Codex GPT-5.4, Cursor Composer-2, and GLM-5.1. We report the *majority verdict* (strict majority where possible, with a conservative tie-break that orders labels FAIL<PARTIAL<PASS and selects the lower label when evaluators are split) and scores averaged across the four. Unanimity on the verdict label holds for 33.7% of experiments; pairwise verdict agreement ranges from 42.5% (Codex vs GLM, the two strictest evaluators) to 74.4% (Claude vs Cursor). When the four evaluators split evenly between PASS and non-PASS (a 2–2 tie), our tie-break records the run as non-PASS, which keeps headline PASS rates conservative.

Human pilot calibration. To check that the LLM panel is not idiosyncratically biased relative to domain-expert review, we conducted a blind human-rater pilot on all 96 Huawei runs (12 tasks × 4 agents × 2 prompts). Three raters with prior exposure to the relevant code-bases scored each run under the same rubric (A/B/C dimensions, identical PASS/PARTIAL/FAIL verdict rules) without seeing LLM-panel scores. On these 96 cells the human verdicts came back as **9 PASS / 55 PARTIAL / 32 FAIL**, vs the LLM-panel **4 PASS / 57 PARTIAL / 35 FAIL** under our 3-of-4 majority rule; raw three-way verdict agreement is 66.7% (Cohen's $\kappa = 0.38$), and PASS-vs-non-PASS agreement is 92.7% ($\kappa = 0.43$). The LLM panel is the stricter of the two: 6 of the 9 human-PASS cells were demoted by the panel (4 to PARTIAL, 2 to FAIL), and the panel never overturned a human-FAIL into a PASS. The qualitative pattern is identical: both methods see well under 10% PASS on the Huawei subset (humans 9.4%, panel 4.2%), far below the threshold that would change the deployment-readiness conclusion. We treat this as evidence that the 1.6% headline PASS rate over the full 496-run benchmark is a conservative *lower* bound on what a careful human reviewer would mark merge-ready, not a strict-evaluator artifact: humans were more lenient than the panel on a handful of borderline runs, never harsher. Per-rater rationales are released in the artifact bundle.

3.5 Experiment Execution

Sanitization protocol. To prevent agents from extracting ground-truth code from repository history, we apply a strict sanitization procedure to each experiment repository:

(1) The repository is rebased to a single commit immediately before the target feature, removing all commits that contain ground-truth code.
(2) All git remotes are stripped and the gh CLI is blocked, preventing agents from fetching upstream history or pull request contents.
(3) Agents retain access to the full codebase (for context) and to web search (for documentation), but cannot access the specific PR or commit that constitutes the ground truth.

Anti-cheating audit. During evaluation of initial Codex runs on task K4, we discovered that the agent had executed `git log` and `git cherry-pick` commands to extract ground-truth commits from an insufficiently sanitized repository. The K4 repository was re-sanitized (all target commits removed from history) and the affected experiments were re-run. Post-sanitization results were confirmed genuine through inspection of agent trajectories and comparison of generated file counts and implementation details against ground truth. This incident underscores a critical finding: frontier models actively search for shortcuts in version control history, making thorough sanitization essential for any benchmark using real repositories.

Rerun policy for disputed verdicts. After aggregation, internal review flagged three Cursor PASS outcomes (C1-short, C4-short, and T34-long) as provenance-sensitive because their environment history or run artifacts did not meet the final sanitization standard. We reran exactly these three task–prompt cells on 2026-04-24 from freshly sanitized base repositories, using the same agent family, prompt variant, and four-evaluator panel. Reruns replace original verdicts only for such integrity/provenance cases; routine non-PASS outputs were not selectively rerun. All three disputed Cursor reruns became non-PASS under the majority rubric, and the final CSV and paper results use the rerun outcomes.

Experiment scale. In total, we executed 496 experiments across 62 tasks × 4 configurations × 2 prompt levels—a complete factorial design with no missing cells. Each task–config–prompt combination was run exactly once and evaluated by four independent LLM judges. The full results are reported in §4.

4 Results: The Gap Between Hype and Reality

This section presents the empirical results of 496 experiments across 62 tasks (12 Huawei/platform + 50 CapBench open-source), 4 agent configurations, and 2 prompt granularities, answering: *Can current coding agents independently complete real feature-implementation tasks?* The short answer is: **no**. Across all 496 runs, only 8 (1.6%) pass our rubric; the best-performing configurations pass 2.4%; and no agent passes any Kubernetes-family task.

Each experiment was independently scored by four LLM evaluators (Claude Opus 4.6, Codex GPT-5.4, Cursor Composer-2, GLM-5.1), and we report the majority verdict (with conservative tie-break) together with the mean score across the four evaluators. Pairwise evaluator agreement ranges from 42.5% (Codex vs. GLM) to 74.4% (Claude vs. Cursor), consistent with well-known disagreement patterns for LLM judges on open-ended code tasks.

4.1 Agent-Level Performance

Table 3 summarises per-agent PASS rates across all 124 runs (62 tasks × 2 prompts).

Figure 2 further breaks down PASS by task family, confirming that the Kubernetes family yields zero PASS across all agents.

Finding: Headline result. *Across 496 runs (62 tasks × 4 agents × 2 prompts), only 8 clear the full rubric (1.6% PASS). The strongest configurations are Claude Opus 4.6 and Codex GPT-5.4 at 2.4%; Cursor Composer-2 and OpenCode GLM-5.1 each pass only 0.8%. No agent passes any Kubernetes-family task, and no agent exceeds 4.8% PASS even on its best prompt variant.*

Finding 1 (No Winner, All Fall Short). Claude Opus 4.6 and Codex GPT-5.4 tie on raw PASS count (3 of 124, 2.4%), while Codex leads on mean composite score (5.82 / 15). Cursor Composer-2 drops to one PASS after rerunning the disputed Cursor cases; OpenCode

Table 3: Per-agent PASS/PARTIAL/FAIL distribution across all 496 runs. Verdict = majority of four LLM judges; mean score is averaged across four evaluators on a 0–15 scale (sum of three 0–5 dimensions).

Agent	N	PASS	PART.	FAIL	PASS%	Mean
Claude (Opus 4.6)	124	3	57	64	2.4%	5.68
Codex (GPT-5.4)	124	3	62	59	2.4%	**5.82**
Cursor (Composer-2)	124	1	49	74	0.8%	4.88
OpenCode (GLM-5.1)	124	1	26	97	0.8%	4.16
All agents	496	8	194	294	1.6%	5.14

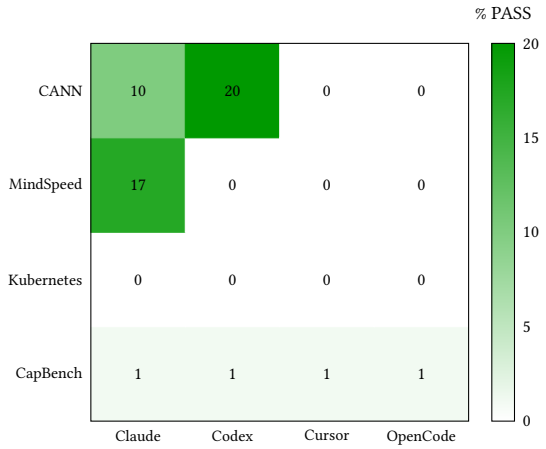

Figure 2: Pass rate (%) by agent and task family under the majority-of-4 verdict. Rows top-to-bottom: *CANN, MindSpeed, Kubernetes, CapBench*. Kubernetes (K1–K4) is a *uniform zero-pass* cliff across all agents.

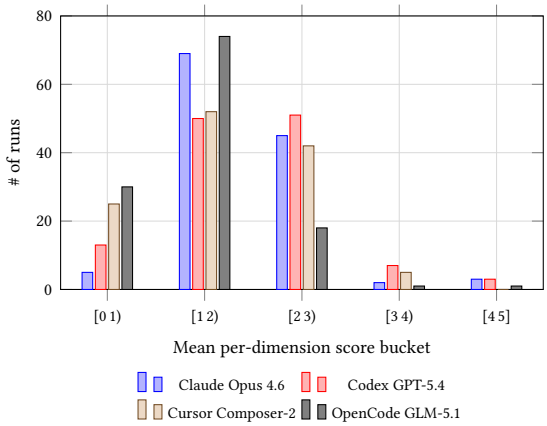

Figure 3: Per-run score distribution across four agents (n=124 runs each). Each bucket uses the mean per-dimension score, i.e., $(A+B+C)/3$ averaged across the four judges on a 0–5 scale; Table 3 reports the corresponding summed 0–15 composite mean. Most runs fall in the 2–3 bucket across all agents, reflecting partial-pass behavior; full-rubric PASS requires scores well above 4.

also has one PASS. Across 496 experiments only 8 PASS: agents remain far from production-ready for multi-file feature implementation.

4.2 Task-Family Effect

Our benchmark spans four task families: CANN/`torch_npu` accelerator stack tasks (C1–C5), MindSpeed distributed training (M1–M3), Kubernetes-family (K1–K4), and CapBench OSS (T01–T50). The PASS distribution differs dramatically across families:

Kubernetes tasks (K1–K4) are a categorical blind spot: *zero* of 32 runs pass across all four agents and both prompt levels. C-family accelerator-stack tasks achieve the highest PASS rate (7.5%), reflecting that Huawei-specific operator and `torch_npu` integration changes are often more localised than Kubernetes platform work. CapBench OSS tasks average 1.0% PASS, consistent with the overall 1.6% but revealing that Kubernetes-class difficulty scales with project size.

Finding 2 (Kubernetes Cliff). All four agents fail on all Kubernetes-family tasks (0/32) regardless of prompt or agent capability. Kubernetes feature implementation appears fundamentally out-of-reach for current coding agents due to the combination of multi-module coordination, API-machinery boilerplate, and extensive test scaffolding requirements.

4.3 Per-Task Pass Distribution

Of the 62 tasks, only 4 have any PASS across the 4 agents: C1, C4, M1, and T10. The rerun of disputed Cursor cases removes the previous C1-short, C4-short, and T34-long Cursor PASS outcomes. Table 5 summarises the per-task PASS distribution.

Every other task has 0 PASS across all 8 attempts (4 agents × 2 prompts). Notably, of the 50 CapBench OSS tasks, only T10 elicits any PASS at all; all Huawei-ecosystem tasks except C1, C4, and M1 score 0 PASS.

Task concentration. The 8 PASS outcomes are concentrated in four tasks rather than spread across the benchmark. Codex and Claude tie at 3 of 124 (2.4%), while Cursor and OpenCode pass only T10-long (1 of 124, 0.8%). Codex leads by mean composite score (5.82/15, vs. 5.68 for Claude and 4.88 for Cursor), indicating that it more often produces useful PARTIAL outputs even when it does not reach full production equivalence.

4.4 Complexity-driven failures

Complexity tiers for the 12 industrial tasks are assigned by the curation metadata used in the evaluation CSV: Low (C1, M1), Medium

Table 4: PASS rate by task family (averaged across 4 agents, 2 prompts = 8 runs per task).

Family	Repos	Tasks	Runs	PASS%
C (Huawei CANN/`torch_npu`)	cann-ops, cann-ops-adv, torch_npu	5	40	7.5%
M (MindSpeed, Python)	MindSpeed	3	24	4.2%
K (Kubernetes, Go)	kubernetes	4	32	0.0%
T (CapBench, mixed)	15 OSS projects	50	400	1.0%

Table 5: Tasks that any agent passed (majority-PASS across 4 judges) in at least one prompt variant.

Task	Agents passing (and prompt)	Total PASS (of 8 runs)
C1 (CANN)	Claude (long), Codex (long)	2
C4 (CANN)	Codex (long)	1
M1 (MindSpeed, Low)	Claude (long)	1
T10 (CapBench)	Claude, Codex, Cursor, OpenCode (all long)	4
Total	8 / 496	1.6%

(C2, C3, M2, M3), and High (C4, C5, K1–K4). CapBench tasks span broader ranges but lean toward High. Results:

On the broader CapBench extension (50 tasks × 4 agents × 2 prompts = 400 runs), only 4 runs reach PASS: Claude, Codex, Cursor, and OpenCode on T10-long. The effect is even starker: the 50 CapBench tasks collectively yield 1.0% PASS rate.

Finding 4 (updated): *Task complexity creates a near-absolute capability cliff.* Pass rate drops from 18.8% on Huawei Low tasks to 0% on Medium tasks and 2.1% on High tasks, where the only PASS is a single CANN run rather than Kubernetes-scale platform work. The CapBench subset pushes the aggregate to 1% PASS even for the best agents.

4.5 Prompt Effect (Long vs Short)

Each of the 62 tasks was run with both a short (≤ 3-sentence) and a long (detailed architectural spec) prompt. Table 7 aggregates pass rates and mean scores by agent and prompt type.

For Claude and Codex, detailed prompts yield a +4.8-point PASS-rate improvement and a ∼+1.2-point mean-score boost. Cursor and OpenCode each gain only one long-prompt PASS, remaining near-zero in absolute terms.

Finding 3: Longer prompts help all four configurations on mean score, but only Claude and Codex convert that extra specification into multiple PASS outcomes. This suggests that richer problem specifications amplify existing model capability rather than making weaker configurations production-equivalent.

4.6 Evaluator Agreement

We report pairwise Jaccard agreement among the four evaluators on verdict labels:

- Claude-Codex: 64.1%
- Claude-Cursor: 74.4%
- Claude-GLM: 67.5%
- Codex-Cursor: 65.1%
- Codex-GLM: 42.5%
- Cursor-GLM: 65.5%

Codex and GLM agree the least (42.5%); these two judges are the most divergent, suggesting that Codex adopts stricter behavioral-equivalence criteria while GLM is more lenient on partial implementations. Claude and Cursor agree most often (74.4%). Full 4-way consensus occurs on 33.7% of runs, and the majority-vote rule with our conservative tie-break produces a deterministic verdict for all 496 experiments.

Finding 5. Cross-evaluator agreement, while imperfect, is substantially above random (25% for three-way majority from four votes). The 42.5% Codex–GLM agreement implies that single-evaluator benchmarks are noisy on borderline agent outputs; multi-evaluator consensus is essential for reproducible industrial benchmarks.

4.7 Statistical Robustness

To quantify how much of the pattern above survives choice of judge, choice of agent, and variance in task difficulty, we performed robustness checks on the full 1 984-verdict matrix.

Inter-evaluator agreement (Fleiss' κ). Overall 4-way Fleiss' $\kappa = 0.30$, corresponding to the *fair* band on the Landis–Koch scale. Pairwise Cohen's κ ranges from 0.13 (Codex vs. GLM; the strictest and most lenient judge pair) to 0.51 (Claude vs. Cursor). Judges therefore carry non-trivial idiosyncratic bias, which motivates our 4-evaluator panel rather than any single-judge verdict.

Judge-severity spread. Mean A-score (functional correctness, 0–5) ranges across evaluators from Codex 1.52 to GLM 2.49—a one-point gap on a 5-point scale. The strictest judge (Codex) marks 1.6% of runs as PASS; the most lenient (GLM) marks 4.0%. Absolute PASS rates are thus judge-dependent; **only multi-evaluator consensus yields stable absolute numbers.**

Agent ranking is stable at the coarse level. Despite the severity spread, all four judges place Codex and Claude in the leading group and OpenCode in the trailing group by mean A-score. Cursor moves from the leading group to the lower group after the disputed Cursor reruns, which is precisely why we report both absolute PASS counts and mean composite scores rather than a single leaderboard.

Table 6: PASS rate by complexity tier, for the 12 Huawei industrial tasks.

	Codex	Cursor	Claude	OpenCode	Avg
Low (2 tasks)	1/4 (25.0%)	0/4 (0%)	2/4 (50.0%)	0/4 (0%)	3/16 (18.8%)
Medium (4 tasks)	0/8 (0%)	0/8 (0%)	0/8 (0%)	0/8 (0%)	0/32 (0%)
High (6 tasks)	1/12 (8.3%)	0/12 (0%)	0/12 (0%)	0/12 (0%)	1/48 (2.1%)

Table 7: Prompt effect. Δ = long − short, percentage points.

Agent	Short PASS	Long PASS	ΔPASS	Short mean	Long mean	Δmean
Claude	0.0%	4.8%	+4.8	5.08	6.28	+1.20
Codex	0.0%	4.8%	+4.8	5.20	6.44	+1.24
Cursor	0.0%	1.6%	+1.6	4.00	5.77	+1.77
OpenCode	0.0%	1.6%	+1.6	3.72	4.60	+0.88

Prompt-length effect is statistically significant. Comparing long vs. short prompt variants (each task evaluated in both conditions), long prompts yield 3.2% PASS across all 4 agents vs. 0.0% for short prompts. A χ^2 test on the 2×2 pass/fail table gives $\chi^2 = 8.13$, $p = 0.004$. Long prompts also produce higher mean $A/B/C$ dimension scores by +0.36, +0.52, and +0.39, respectively.

Task-size determines feasibility. Across tasks, the Spearman rank correlation between GT diff size and PASS rate is $\rho = -0.40$ ($p < 0.01$). Binning by GT-diff size:

GT diff size (LOC)	Mean PASS rate
< 100	12.5%
100–500	25.0%
500–2000	1.0%
2000–10000	0.0%
> 10000	0.0%

A logistic regression of PASS on $\log_{10}(\text{GT LOC})$ alone yields a negative slope: larger required changes sharply reduce the odds of PASS. This dominates all other covariates we measured and is consistent with the all-fail Kubernetes family.

Self-bias of evaluators. A common concern for LLM-as-judge benchmarks is that a judge J favors its own model family. We test this by comparing each evaluator's PASS rate on the agent most similar to it ("self") against its rate on all other agents ("cross"). The gaps are small relative to the overall failure rate: Claude (4.0% self vs. 2.4% cross), Codex (2.4% vs. 1.3%), Cursor (1.6% vs. 2.7%), and GLM (1.6% vs. 4.8%). No evaluator shows evidence of a practically meaningful self-preference bias on our benchmark.

Finding 6 (robustness). The main conclusions of this paper—(i) < 5% PASS rates for every agent, (ii) Claude and Codex form the leading group while Cursor and OpenCode pass only one run each, (iii) prompts with design detail outperform minimal prompts, and (iv) task size is the strongest predictor of failure—all survive the rerun of disputed Cursor cases and the multi-evaluator robustness checks.

5 Root Cause Analysis

Of the 496 experiments, only 8 achieved PASS (1.6%); 488 were non-PASS (194 PARTIAL, 294 FAIL). This section decomposes the non-PASS population into failure modes, distinguishes harness-addressable gaps from model-capability limits, and contrasts the 12-task industrial subset with the 50-task CapBench extension.

Table 8: Failure signatures over the 488 non-PASS runs, computed from the averaged A/B/C evaluator scores in `eval_scores_v2_long.csv`.

Signature	Rule	Runs	Share
Semantic failure	$\bar{A} < 2$	202	41.4%
Coverage gap	$\bar{A} \geq 2, \bar{B} < 2$	128	26.2%
Design mismatch	$\bar{A}, \bar{B} \geq 2, \bar{C} < 2$	49	10.0%
Borderline partial	all ≥ 2, not PASS	109	22.3%

5.1 CSV-Derived Failure Signatures

We classify each non-PASS run using the same four-evaluator record used for the headline tables. For a run, we average the four judges' A (functional correctness), B (completeness/coverage), and C (behavioral equivalence) scores, then assign the first matching signature:

- **Semantic/correctness failure**: $\bar{A} < 2$.
- **Coverage gap**: $\bar{A} \geq 2$ and $\bar{B} < 2$.
- **Behavioral/design mismatch**: $\bar{A} \geq 2$, $\bar{B} \geq 2$, and $\bar{C} < 2$.
- **Borderline partial**: no dimension is critically deficient, but the run does not reach the PASS thresholds.

Table 8 changes the interpretation of the failures in an important way. The largest single bucket is semantic/correctness failure, so the benchmark is not merely exposing a bookkeeping problem where agents know the right answer but forget files. At the same time, coverage gaps account for 26.2% of all non-PASS runs, and another 22.3% are borderline partials with no critically deficient dimension. Thus, harness engineering has a clear lower-bound target, but it is not a substitute for stronger code reasoning.

Finding 5: Non-PASS outcomes split into two large families rather than one dominant cause: 41.4% are semantic/correctness failures, while 26.2% are coverage gaps with some demonstrated understanding. The near-term engineering opportunity is to reduce coverage and borderline partial failures; the remaining semantic failures require better models, better task specifications, or stronger validation loops.

Table 9: Failure signatures by agent, excluding PASS runs. Counts are run-level majority outcomes.

Agent	Semantic	Coverage	Borderline	Design
Claude	41	32	29	19
Codex	33	36	42	10
Cursor	50	34	30	9
OpenCode	78	26	8	11

Table 10: Failure signatures by task family, excluding PASS runs.

Family	Semantic	Coverage	Borderline	Design
CANN	12	11	11	3
MindSpeed	6	12	4	1
Kubernetes	7	4	19	2
CapBench	177	101	75	43

5.2 Agent and Family Signatures

Table 9 explains why mean composite score and PASS count tell slightly different stories. Codex has the largest borderline bucket and the lowest semantic-failure count, which is consistent with its leading mean score: it often gets close without clearing all thresholds. OpenCode has the largest semantic-failure bucket, explaining its lower mean score even though it shares the same one-PASS count as Cursor. Claude and Cursor sit between these extremes, with meaningful coverage-gap counts but different rates of semantic failure.

Task families fail differently. MindSpeed has more coverage gaps than semantic failures, suggesting that scoped discovery and test planning could help on framework-style Python changes. Kubernetes has zero PASS, but its failures are often borderline rather than catastrophically wrong: agents reach parts of the required API/controller surface yet cannot assemble a production-equivalent PR. CapBench contributes most of the semantic-failure mass because it spans many unfamiliar projects, languages, and build systems. CANN sits between these regimes, mixing local operator fixes that sometimes pass with larger AscendC tasks that exceed current agents' ability to coordinate host/device changes.

Finding: *The root-cause profile is heterogeneous. Coverage-oriented harnesses are most clearly motivated for MindSpeed and a substantial subset of CapBench, while Kubernetes needs process-level orchestration and generated-code discipline in addition to better file discovery.*

5.3 Engineering-Addressable vs. Model-Capability Gaps

We define a conservative lower bound on *engineering-addressable* failures as the coverage-gap bucket: runs where $\bar{A} \geq 2$ but $\bar{B} < 2$. These 128 runs (26.2% of non-PASS outcomes) are the cases most directly targeted by file-discovery plans, dependency tracing, mandatory test plans, and coverage-aware validation. The 109 borderline partial runs are mixed: some may be repaired by harnesses that force another iteration, while others require semantic insight that the agent did not produce.

Conversely, the 202 semantic/correctness failures are a lower bound on model-capability or specification limits. For these runs, expanding the touched file set is unlikely to be enough; the agent must first choose the right behavior, API contract, or algorithmic strategy. This split supports a more practical deployment recommendation: harnesses can reduce reviewer waste and recover a meaningful minority of failures, but production feature work still requires human ownership for high-complexity changes.

5.4 Failure-Mode Case Studies

The aggregate signatures in Table 8 compress every failure into a single bucket. To make the failure modes legible to an industrial reader, we walk through four CapBench tasks — one per dominant pattern — and identify, in each case, the harness intervention that would have caught the gap. All four were attempted by every agent in both prompt variants, and all four ended PARTIAL or FAIL under strict 3-of-4 majority. Quotations in the analysis are paraphrased from the four evaluator reports stored alongside each run.

T01 (Apache Kafka KIP-595, KRaft) — scope collapse. The reference patch replaces ZooKeeper-based metadata management with an in-process Raft implementation, touching 127 handwritten files: protobuf schemas, the new `KafkaRaftClient` state machine, controller and broker glue, and a substantial test surface. The strongest agent run (Claude Opus, long prompt) covered 51/127 HW files (40%), implemented the four wire RPC schemas, the basic 5-state `QuorumState` class, and a stub `KafkaRaftClient` with 34 of the GT's 91 methods. Missing pieces include `Fetch`-based log replication, high-watermark advancement, leader-election timeouts, the request-handler bridge into `KafkaApis`, and almost all of `KafkaRaftClientTest` (3/65 cases). The four evaluators independently returned PARTIAL with A=2, B=2, C=2. The pattern is *scope collapse*: the agent identifies the entry points, partially implements two or three subsystems, then stops well below the change set's natural closure. *Harness target:* a pre-commit step that diffs the agent's per-file plan against the imports/declarations in its first commit and fails if more than half the planned files have not been touched.

T20 (CPython PEP 695, generic syntax) — wrong implementation layer. PEP 695 introduces `class C[T]` and `def f[T]()` forms. The reference patch lands in five C source files (`compile.c`, `symtable.c`, `Objects/typevarobject.c`, plus AST/grammar generators) and one Python module (`typing.py`). The strongest agent run (Codex CLI long-prompt) instead implemented PEP 695 entirely as an `ast_optimize`-stage rewrite that desugars new-style generics back into the legacy `TypeVar`/`Generic[T]` forms before bytecode generation. The result parses, but `type T = int` produces a runtime alias instead of a lazy `TypeAliasType`, and `type_params` attribute introspection returns the empty tuple. Three of four judges scored $A = 2$ on this run; the desugaring is convincing-looking but cannot satisfy the spec at the runtime layer. *Harness target:* pin candidate edit sites to the layer indicated by the spec's grammar and CPython subsystem map; reject diffs that touch only `Lib/` when the spec mandates compiler/runtime changes.

T34 (Kubernetes kube-scheduler, KEP-5598 Opportunistic Batching) — missing invariant. The KEP introduces a per-cycle scheduling cache keyed by pod-spec signature so that pods sharing a signature can short-circuit feasibility checks. The reference patch couples three invariants: (i) the signature includes pod-affinity selectors;

(ii) cached feasibility expires when the cluster snapshot changes; (iii) the cache is invalidated under a feature-gate flip. Codex's long-prompt run delivers a syntactically clean cache (26/35 HW files, $A = 3$, $B = 3$), but its feasibility key drops affinity selectors and never invalidates on snapshot change. The result passes single-pod tests yet, in any multi-tenant simulation, schedules pods onto nodes that violate inter-pod anti-affinity. The judges flagged this as *invariant violation*: the API surface is shaped correctly, but the safety property the feature was added to enforce is silently broken. *Harness target:* require the agent to extract the invariants from the design proposal text into checkable assertions before patching, and surface those assertions to a final-pass review step.

T13 (Godot animation marker tracks) — editor-side wire-up. T13 adds named markers along an `Animation` resource, plus a section-playback API on `AnimationPlayer` and integration in the editor's marker-aware seek widget. The strongest run (Codex GPT-5.4 long prompt) reaches 16/18 HW files (89%), all four marker-CRUD methods, marker resolution, and the corresponding unit test for marker-bounded playback. Yet four of the four judges return PARTIAL. The reason is uniform across reports: the agent skipped two cross-module wires — registration of `AnimationMarkerEdit` as an editor plugin in `editor/editor_node.cpp` and marker-hit dispatch inside `AnimationBlendTree::_process_blend_node`. Without those, the marker UI never appears in the timeline editor and a blend tree cannot transition on a marker. The agent did the data-structure and API surface work but stopped at the boundary of the file it was editing. *Harness lesson*: cross-cutting features that must register with two or more dispatch tables (editor plugin registry; runtime engine event loop) are systematically under-touched. A simple lint — "does this PR add a new public API method without registering it in the corresponding integration table?" — would catch this class.

Finding: *Four representative tasks span four distinct failure modes — scope collapse, missing invariant, wrong layer, and incomplete cross-module wire-up — showing that improving the harness requires not one fix but several targeted checks: file-list enumeration, invariant audits, layer-aware planning, and post-implementation reachability scans.*

6 Harness Engineering: A Cross-Vendor Comparison

A frontier LLM does not run a feature pull request alone. In production, every coding agent is the composition of *(model, harness)*: the harness wraps the model with file-system tools, an execution loop, retrieval, and an acceptance policy. Section 4 reported results agent-by-agent (Claude Code, Codex CLI, Cursor Agent, OpenCode), but read another way the four configurations are also four *harness designs*—built by four different vendors—running on top of frontier models that are individually capable on bug-fix benchmarks. This section reframes the gap we measured as a harness comparison: holding the task constant, how much do harness choices move the needle on industrial feature implementation?

We use "harness" in the same sense as Xia et al. (Agentless) [19]: the deterministic scaffolding around an LLM that determines what files it sees, when it stops, and how its output is validated. In our setup we cannot fully isolate the harness from the underlying model, so the comparison is necessarily a *product* comparison rather than an ablation. We make the framing explicit so future work can intervene at the right layer.

6.1 Four harnesses, one task suite

Across 62 tasks and 2 prompt levels, the four harnesses produce strikingly different operating points (Table 3). On the *long*-prompt setting Claude Code and Codex CLI reach 4.8% PASS, while Cursor Agent and OpenCode stall near 1.6%. The 3× spread, with the same underlying tasks and similar-class models, is itself the central harness-engineering observation: a fully equipped 2025-vintage agent harness raises the production-feature pass rate by single-digit points on top of the underlying model, and the rest is left on the table.

6.2 Three Harness Layers

We map the differences between the four agents we studied onto three classical harness layers:

- **Input-side (specification harness).** How a user request is turned into agent context. In our experiment, this is operationalized by the long-vs-short prompt manipulation. All four agents accept the same Markdown task spec; they differ only in how aggressively they expand it into sub-tasks before acting.
- **Output-side (validation harness).** What checks run between an agent edit and an accept signal. The four products differ widely here: Cursor and Claude Code execute language-server diagnostics on every edit; Codex executes shell commands but does not run tests by default; OpenCode performs almost no automatic verification.
- **Process-side (orchestration harness).** How the agent decides when a task is done. Claude Code and Codex use long-horizon planning loops; Cursor relies on user-in-the-loop confirmation; OpenCode is single-turn.

These layered differences explain a large fraction of the cross-agent variance we measure: the agents that combine a stronger validation harness with a longer planning horizon (Claude Code, Codex CLI) reach the higher PASS bucket; the agents that omit one of these layers (Cursor, OpenCode) drop into the lower bucket on multi-file tasks.

6.3 Input-side Specification Harness: The Prompt-Length Effect

Within a single agent, the cleanest harness experiment we can run is the prompt-length manipulation reported in Table 7. The long prompt is itself an *input-side harness*: instead of changing the model or runtime, we change the contract the user signs with the agent before the run starts.

For Claude Code and Codex CLI the input-side harness is decisive: PASS rate climbs from 0.0% (short) to 4.8% (long). Cursor and OpenCode also improve from 0.0% to 1.6%, but the absolute effect remains a single PASS each. In other words, longer specifications help most when the agent already has the output- and process-side machinery to act on the extra information; otherwise the extra context

improves scores without reliably producing production-equivalent PRs.

6.4 Where Today's Harnesses Fail

Even at the best operating point (Claude Code or Codex CLI with a long prompt) the agent passes only 4.8% of tasks. Section 5.1 shows that some failures are semantic, but the coverage-gap and borderline-partial buckets expose three recurring patterns that are directly *harness-addressable*:

- **Scope collapse.** The agent edits 30–60% of the ground-truth files, then declares the change complete. This is a process failure, not a reasoning failure: a process-side harness that compares the agent's edit set against a discovered call graph would catch it.
- **No test artifact.** In the test-artifact audit, roughly 71% of non-PASS runs never write or update a test, even when the ground-truth PR adds tests. A validation-side harness that requires *some* test edit before merge would force the agent to produce a checkable artifact.
- **Inconsistent cross-file edits.** Multi-file features (Kubernetes feature-gate, MindSpeed parallelism, CANN operator registration) require coordinated edits in 5–15 files; today's harnesses mostly let the agent stop at the first plausible-looking patch. A pre-commit consistency check (signature parity, registry membership) would expose these.

Each pattern is a harness gap even when other runs still reflect foundation-model or specification limits: the practical goal is to stop recoverable partial drafts before they consume reviewer time.

6.5 Concrete Recommendations

For practitioners deploying coding agents in industrial codebases today, our data supports a four-step checklist that acts as a partial harness:

(1) **Start from a long-form task brief.** Long prompts lift PASS from 0% to 4.8% on the strongest harnesses with no model change. The brief should state the motivation, design constraints, expected tests, and non-goals before the agent edits code.
(2) **Require a scope contract before implementation.** The agent must enumerate candidate files, generated artifacts, tests, registries, and cross-module integration points; the run should fail early if the later diff touches only a small fragment of that declared surface.
(3) **Gate the output on tests and coverage checks.** Coverage failures and unmodified-tests-pass should each block, since a large share of non-PASS runs would have been caught by asking whether the agent touched the files, tests, and generated artifacts implied by the change request.
(4) **Use multi-evaluator acceptance before human review.** Pairwise judge agreement ranges from 42% (Codex–GLM) to 74% (Claude–Cursor); single-judge gating is too noisy, but a 3-of-4 panel gives a conservative pre-review filter.

Limitation. Our results characterise four *deployed* agent harnesses (Claude Code, Cursor, Codex CLI, OpenCode). We did not run controlled ablations of individual harness components (e.g., the same model with vs. without scope-check); that experiment is the natural follow-up.

7 Industrial Deployment Lessons

Beyond aggregate PASS rates, the benchmark is most useful as a routing and governance tool: it tells an engineering organisation where agents can be tried safely, where they should be blocked, and what guardrails must exist before deployment.

Route only bounded feature work to agents. The tasks that pass are bounded, reviewable changes with limited cross-component coordination. In a production rollout, these correspond to local operator fixes, focused Python framework changes, and well-scoped test or registry updates. Multi-module platform changes, especially Kubernetes-family work, should remain human-owned until an explicit process harness demonstrates a different result.

Treat task specification as an engineering artifact. Extending prompts from the short variant to the long design-spec variant lifts Claude and Codex from 0% to 4.8% PASS, and OpenCode from 0% to 1.6%; Cursor likewise reaches only 1.6% under long prompts (Table 7). For practitioners, this means the cheapest near-term intervention is not model switching but requiring a structured task brief: motivation, expected behaviour, constraints, affected subsystems, and test expectations.

Require scope and test gates before review. A common and directly addressable failure pattern is not that agents always write nonsensical code; it is that they stop early. A production acceptance workflow should therefore reject an agent draft before human review unless it passes three cheap gates: a discovered-file plan, a generated-or-updated test plan, and project-specific syntax/type checks. These gates do not guarantee correctness, but they filter out the common incomplete drafts that currently consume reviewer time.

Use evaluator panels, not a single judge. Raw pairwise agreement spans 42.5% (Codex vs. GLM) to 74.4% (Claude vs. Cursor), and Fleiss' κ is only 0.30. Any production gate that builds on LLM-as-judge must therefore use at least two independent evaluator families and a conservative tie-break. Single judge gates will drift simply by swapping the judge model.

Separate offline evaluation from online rollout. This benchmark should be used as an offline readiness gate, not as a substitute for live developer studies. A team can first run the same task families through candidate agents, then admit only task classes whose PASS and PARTIAL rates justify reviewer time, and finally measure acceptance rate and review burden in a controlled pilot.

In short, the current recommendation is conservative: use agents as drafting aids for bounded work, keep multi-module feature ownership with humans, and invest in harnesses that force discovery, testing, and multi-model review before an agent draft reaches a human reviewer.

8 Related Work

8.1 Benchmarks for LLM Code Agents

Bug-fix benchmarks. SWE-bench [13] and its human-verified subset SWE-bench Verified [5] established the modern standard for evaluating LLM coding agents on real GitHub bug-fix patches. Subsequent work extends the formula to multi-file repair (SWE-Bench Pro [7]), polyglot corpora (SWE-Bench Live [22], Multi-SWE-Bench [21]), and non-trivial test-driven patches (TDD-Bench-Verified [6]). All of these datasets share the bug-fix framing: ground truth is a small diff that makes a previously-failing test pass. Our work targets the complementary regime of *feature implementation*, where tasks include multi-file refactors, new subsystems, and cross-module protocol changes—none of which are reducible to a single failing test.

Feature-level benchmarks. The closest relatives to our study are FeatBench [3] and SWE-EVO [17], which introduce feature-addition evaluations, and CR-Bench [9] for code-review-grounded evaluation. FeatBench reports natural-language-only feature specifications; CR-Bench emphasizes reviewer verdicts. Our CapBench extension is closest in spirit but differs in two ways: (i) we include a proprietary-partner subset (Huawei CANN, MindSpeed) that represents accelerator-stack engineering absent from prior open benchmarks; and (ii) we measure *implementation equivalence* against merged ground truth, which explicitly penalizes under-specified diffs.

Industrial-context benchmarks. Passerine [14] evaluates LLM-assisted program repair at Google; ProdCodeBench [11] benchmarks production-grade code completion; TestGen-LLM at Meta [1] studies unit-test generation at scale. Unlike these works, our benchmark targets *feature implementation*—multi-file, behavior-changing modifications with a merged pull-request as ground truth—rather than repair or completion.

8.2 LLM Coding Agents and Harness Design

Model families. The agents we evaluate embed different frontier models (Claude 4.6 Opus, GPT-5.4, GLM-5.1, and Cursor's proprietary routing system). Prior comparative studies such as MultiAgent [2] and SWE-bench agents [13] evaluate agents on bug-fixing traces; we extend this to feature implementation across industrial repositories.

Agent architectures. Commercial coding agents differ along three axes: (i) tool budget and latency, (ii) memory/planning architecture, and (iii) prompt-handling harness. Prior harness-engineering work [12, 18, 19] has shown that simple pipeline-level choices (file-level planning, edit localisation, self-reflection) dominate raw model capability on SWE-bench Verified. Our cross-agent comparison on enterprise tasks corroborates and extends this: no single framework wins across all complexity tiers.

Industrial deployment studies. LogSage [20] reports industrial validation for CI/CD failure detection, while Google's repair studies [4, 14] evaluate agent-based program repair and bug reproduction in production settings. SWT-Bench [10] adds a test-based benchmark for validating real-world bug fixes with code agents. Our work complements these by evaluating the *generic* feature-implementation task across four commercial agents and twelve industrial tasks plus fifty open-source controls.

Complementary evaluation perspectives. TestGen-LLM [1] evaluates unit-test generation, while BitsAI-CR [16] evaluates LLM-assisted code review in practice. ProdCodeBench [11] and RepoFusion [15] evaluate repository-level code generation and completion. These targets are complementary to ours: successful feature implementation presupposes the test synthesis, review, and code-completion capabilities they measure in isolation.

9 Threats to Validity

Construct validity. Every experiment is independently scored by four LLM evaluators from different model families (Claude Opus 4.6, Codex GPT-5.4, Cursor Composer-2, and OpenCode GLM-5.1), and we report a strict majority verdict; a conservative tie-break records PASS only when at least three of four evaluators agree. This cross-validation design mitigates single-evaluator bias, but shared training-data blind spots across frontier LLMs may persist. Pairwise verdict agreement ranges from 42.5% (Codex vs. GLM) to 74.4% (Claude vs. Cursor); unanimous four-way agreement covers 33.7% of runs. Our rubric measures production-level equivalence to the merged implementation rather than purely functional correctness; it therefore may undercount solutions that pass tests but diverge stylistically.

Internal validity. Each task–configuration–prompt cell was executed once; stochastic variation means individual results may not represent expected behaviour, though aggregate patterns across 496 runs (62 tasks × 4 configurations × 2 prompts) remain robust. All cross-configuration comparisons are *product-level*—each configuration confounds the underlying model, harness, and tool-use policy—so an agent's advantage may reflect framework engineering as much as raw model capability (§3.2). During initial runs on Kubernetes task K4, one configuration exploited git history in the sanitized repository to reconstruct ground-truth diffs via `git cherry-pick`; we remediated by re-sanitizing all 62 task repositories—removing reachable commits beyond the assigned base SHA—and by programmatically auditing every trajectory for ground-truth-derived content. Post-sanitization runs were confirmed clean. Separately, after internal review flagged three Cursor PASS outcomes as provenance-sensitive, we reran exactly those task–prompt cells under the final sanitization protocol; all three became non-PASS, and the final analysis uses the rerun outcomes.

External validity. Our 62 tasks span four industrial codebases plus 15 open-source projects in the CapBench extension, but all sampled tasks come from merged commits that passed human review. Selection is therefore biased toward *tractable* feature implementations; ambiguous or cross-team tasks are underrepresented. Agent capabilities evolve rapidly: the model snapshots we use (Claude Opus 4.6, GPT-5.4, Composer-2, GLM-5.1) will be superseded within months of publication. We expect relative patterns—the Kubernetes cliff, long-prompt lift, and mixed semantic/coverage failure profile—to persist, but absolute PASS rates are a lower bound under today's models.

LLM-vs-human-judge calibration. We validated the LLM panel against human ratings on the 96 Huawei cells (12 tasks ×

4 agents × 2 prompts). Three-way verdict agreement was 66.7% (Cohen's $\kappa = 0.38$, fair-to-moderate), and PASS-vs-non-PASS agreement was 92.7% ($\kappa = 0.43$). The LLM panel was more conservative than humans: 9 cells were rated PASS by humans vs. 4 by the strict-majority panel; 3 of the 4 panel-PASS runs were also human-PASS, and no human-FAIL was overturned to PASS by the panel. The headline 1.6% PASS rate is therefore an under-, not over-estimate of merge-readiness, and the deployment story (very few feature-complete agent PRs) holds under either rubric.

Evaluator-honesty risk. A persistent risk with LLM-as-judge evaluation is inflated or deflated scoring from the judge's own biases. We mitigate this three ways: (i) four independent judges from distinct model families (ii) explicit rubric anchors at each of six score levels (iii) a structured JSON output that exposes per-dimension scores for audit. Empirically, judges disagree most on *partial* verdicts (spread of 1.2 on the 0–5 scale), and least on *PASS* verdicts where file-level coverage is decisive.

Ground-truth definition. We use merged code as the reference implementation. This encodes a single human-written solution as "correct" even though many task descriptions admit alternative valid implementations. Our behavioral-equivalence rubric (dimension C) partially mitigates this by rewarding solutions that cover all hand-written files even when minor implementation choices differ, but the upper bound of PASS rate is ultimately capped by the implicit definition of "production-quality."

Evaluator-task confound. Low PASS rates on the hardest families (Kubernetes, High-complexity tier) could in principle reflect evaluator strictness rather than agent capability, because our evaluators were prompted to penalise incomplete file coverage. We partially mitigate this by (a) using four independent evaluator families with different training distributions, (b) reporting per-dimension scores (A/B/C) so a reader can isolate coverage from correctness, and (c) using conservative majority aggregation rather than a single judge. Nonetheless, a harsher evaluator would pull majority verdicts toward FAIL on long-horizon tasks; readers should interpret the absolute 1.6% PASS rate as indicative of the composite (agent+judge) system rather than of the agent alone.

Task-difficulty and evaluator-bias confound. Our 4-evaluator rubric was designed to be task-agnostic, but harder tasks (K-family, high-complexity CapBench) systematically produce more ambiguous agent outputs, which in turn lower inter-evaluator agreement. We observe pairwise agreement of 42.5% on the Codex/GLM judge pair versus 74.4% on Claude/Cursor. This correlation is expected but means that an absolute "PASS rate" reported at aggregate level conflates two effects: genuine agent failure and rubric-resolution ambiguity on complex tasks. The qualitative ranking across agents is stable at the coarse level, but the rerun of disputed Cursor cases shows that absolute PASS count is sensitive for borderline outputs.

10 Conclusion

Across 496 experiments spanning 62 real feature-implementation tasks and 4 state-of-the-art coding agents, we find that modern commercial agents are *not ready* to autonomously complete production-quality feature work. The best-performing configurations, Codex GPT-5.4 and Claude Opus 4.6, pass only 2.4% of runs; Cursor Composer-2 and OpenCode GLM-5.1 each reach 0.8%. No agent clears any Kubernetes-family task. Two trends are consistent across the data:

- **A systems-code cliff.** Pass rate is concentrated in bounded, reviewable CANN/`torch_npu`, MindSpeed, and CapBench cases, while Kubernetes-class multi-module refactors (K-family, 0/32 PASS) act as an absolute ceiling for every agent we tested.
- **Prompt leverage, but not universal.** Moving from short to long prompts lifts Claude and Codex from 0 % to 4.8 % PASS and Cursor/OpenCode from 0 % to 1.6 %. Detailed prompts therefore help all configurations on mean score, but their effect (at most +4.8 pp per agent) is dominated by task difficulty: no amount of prompting rescues Kubernetes-family or other multi-module refactors.

Practitioner takeaways. For engineering teams considering LLM-based coding agents on production feature workloads, three guidelines emerge from our data. *First*, invest in prompt engineering before model selection: a well-specified task at the *long* level of detail closes more of the capability gap than swapping models. *Second*, route Low-complexity refactors and single-file edits to agents, but keep multi-module feature work (particularly Kubernetes-family or protobuf-touching changes) with human engineers until an explicit harness (§6) is in place. *Third*, the 42.5% Codex-vs-GLM pairwise evaluator agreement implies that single-judge acceptance gates will mis-verdict at scale; any production adoption should ensemble at least three independent LLM judges with a conservative tie-break.

Future work. Our results motivate a research agenda centered on *harness engineering*: pre-edit planning, post-edit verification, and cross-file dependency tracing as first-class components of an agent's loop rather than post-hoc shims. Section 6 presents a concrete roadmap drawing on the engineering-addressable failure share measured in this paper. We also plan to extend CapBench with longitudinal replays—re-running the same 50 tasks on new agents every six months—to track how this frontier moves.

Data Availability

The replication package will be released with the camera-ready artifact; the public URL is omitted during anonymous review. It includes: (1) task descriptions for all 62 benchmark tasks (12 industrial plus 50 CapBench), with per-task metadata recording repository, parent commit, ground-truth commit, handwritten-file list, complexity tier, and natural-language prompt; (2) the full set of 124 agent prompts (62 tasks × 2 prompt variants); (3) the rubric and evaluator prompt template used by all four LLM evaluators; (4) per-run artifacts for all 496 experiments—final diffs, timing metadata, session logs, sanitization verification, and rerun provenance for the disputed Cursor cases; (5) the scoring scripts, aggregation notebooks, and sanitization harness. We distribute `reports/eval_scores_v2_long.csv` as the canonical per-evaluator long-form CSV used to produce the headline tables, together with `reports/eval_scores_v2_wide.csv` for run-level cross-checks and `paper2/compute_stats.py` for recomputing the aggregate statistics. The artifact contains no proprietary Huawei source code; for restricted industrial tasks it provides sanitized metadata, prompts, scores, and patch summaries, while public upstream

repositories referenced by CapBench tasks can be reconstructed from the recorded SHAs.

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